

Script-Controlled ACL – Restrict Record Access Based on Field Value

Phase 5 – Project Development

1. Create a User

1. Log in to ServiceNow as Admin
2. Go to **User Administration** → **Users**
3. Click **New**
4. Fill in:

User ID: EEE User

First name: EEE

Last name: User

Email: eeeuser@gmail.com

1. Click **Save**

Script-Controlled ACL-Micro P... x ServiceNow Developers x New Record | User | ServiceNow x + Ask Gemini

dev446707.service-now.com/now/nav/ui/classic/params/target/sys_user.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_user%26sysparm_che...

servicenow All Favorites History Workspaces User - New Record Search

User New record Submit

To set up the User's password, save the record and then click Set Password.

User ID EEE User

First name EEE

Last name User

Title

Department

Password needs reset

Locked out

Active

Identity type Human

Internal Integration User

Email eeeuser@gmail.com

Language -- None --

Calendar integration Outlook

Time zone System (America/Los_Angeles)

Date format System (2026-08-19)

Business phone

Mobile phone

Photo Click to add...

Submit

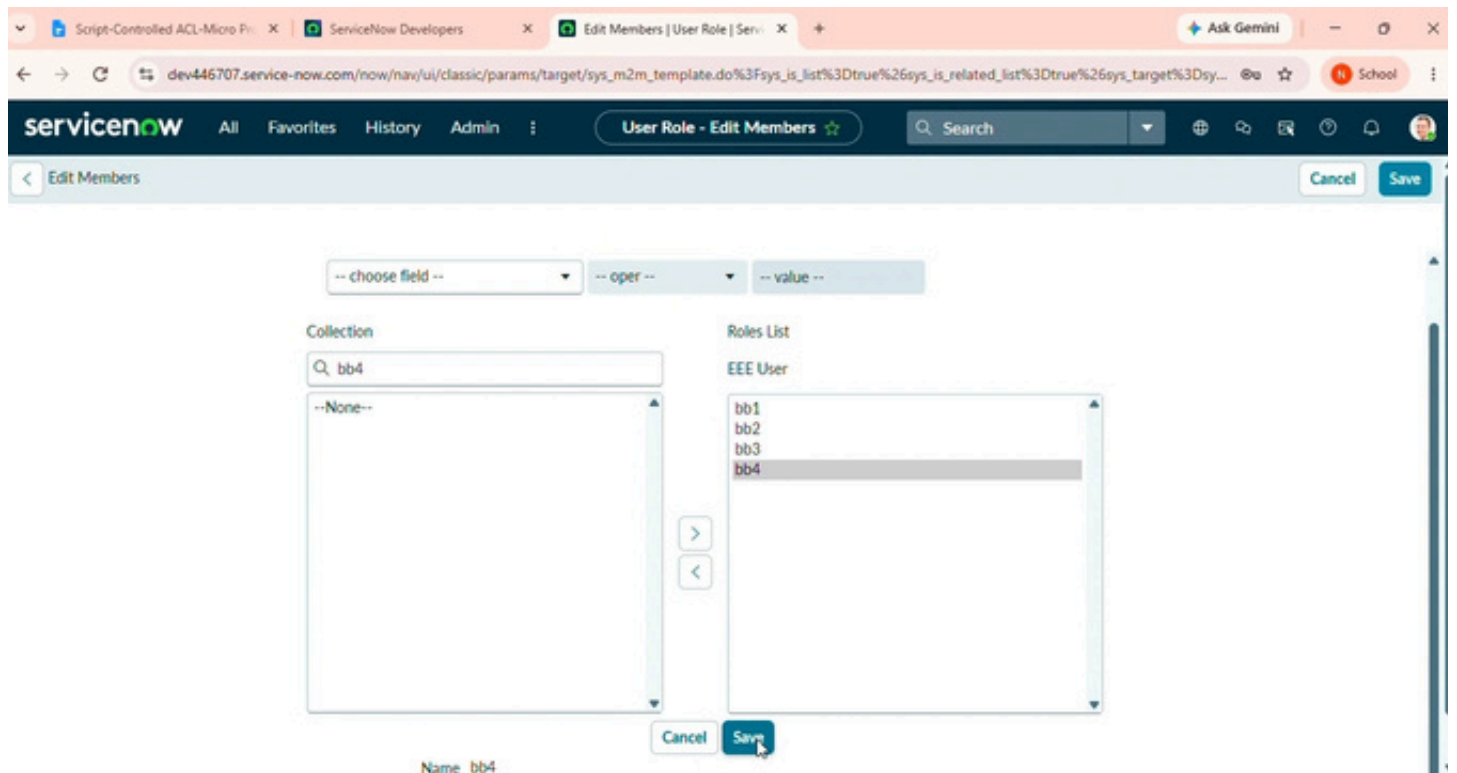
2. Create Roles

1. Go to **User Administration** → **Roles**

2. Click **New**, type role name bb1, click **Submit**
3. Repeat and create bb2, bb3, bb4 the same way

The screenshot shows the ServiceNow 'Role - New Record' form. The browser address bar indicates the URL is `dev446707.service-now.com/now/nav/ui/classic/params/target/sys_user_role.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_user_role%26sys...`. The ServiceNow header includes the logo, navigation tabs (All, Favorites, History, Workspaces), a search bar, and a 'Role - New Record' button. The form fields are: 'Name' (required, value 'bb1'), 'Application' (dropdown, value 'Global'), 'Elevated privilege' (checkbox, unchecked), and 'Description' (text area). A 'Submit' button is located at the bottom left of the form area.

1. Open EEEUser and add all 4 roles to this user



3. Create the Table

1. Go to **System Definition** → **Tables**
2. Click **New**
3. Fill in:
 - Label: Institution Details
 - Name: u_institution_details
 - Extends: leave as false

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More info](#)

* Label: Institution Details

* Name: u_institution_details

Extends table: [Search]

Application: Global

Create module: ☒

Create mobile module: ☒

Add module to menu: -- Create new --

New menu name: Institution Details

Remote Table: ☐

Columns | Controls | Application Access

Table Columns: for text [Search]

Dictionary Entries

1. Add these fields:
 - Student Roll Number – Auto Number
 - Student Name – Reference (User)
 - Faculty Name – Reference (User)
 - Branch – Choice (ECE, EEE, CSE)
 - Email – String
 - Phone Number – String
 - Description – Multi-line String
2. Save and add some sample records with different branches

Table - Institution Details

Field	Type	Value	Length	Required
Updated by	String	(empty)	40	false
Updated	Date/Time	(empty)	40	false
Sys ID	Sys ID (GUID)	(empty)	32	false
Created by	String	(empty)	40	false
Created	Date/Time	(empty)	40	false
Student Roll Number	String			false
Student Name	Reference	User		false

Buttons: Delete, Update, Delete All Records

Related Links: Form Builder, Design Form, Layout Form, Layout List, Show Form, Show List, Show Schema Map, Add to Service Catalog, Run Point Scan, Explore REST API

Institution Details

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
ECE			Abraham Lincoln		Abel Tuter	STU100
CSE			Andrew Och		Amos Linnan	STU300
EEE			Alfonso Griglen		Alejandra Prenatt	STU200

4. Create the Read ACL (Main Script)

1. Go to **System Security** → **Access Control (ACL)**
2. First, elevate your role: click your profile picture → **Elevate Role** → tick `security_admin`

3. Click **New**
4. Fill in:
 - Type: record
 - Operation: read
 - Name: u_institution_details
 - Active: true
 - Advanced: true (this shows the script box)
5. In **Requires role**, add bb1
6. In **Data Condition**, set Branch is EEE
7. In the **Script** box, type

```
(function () {  
  
    // Allow admin users full access  
  
    if (gs.hasRole('admin')) {  
  
        return true;  
  
    }  
  
    // Allow only EEE branch users to see EEE records  
  
    if (gs.hasRole('bb1')){  
  
        return true;  
  
    }  
  
    // Deny access for all others  
  
    return false;  
  
    })());
```

8. Click Submit

Script-Controlled ACL-Micro P... ServiceNow Developers New Record | Access Control |

dev446707.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_security_acl%26sysp...

ServiceNow All Favorites History Admin Access Control - New Record Search

Access Control New record

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type record Application Global

* Operation read Active ☒

Admin overrides ☒ Advanced ☐

Decision Type Allow If

Protection policy -- None -- Tables fields

* Name Institution Details [u_institution_details] -- None --

Description

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[Avoid\(\)](#)

Script-Controlled ACL-Micro P... ServiceNow Developers New Record | Access Control |

dev446707.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_security_acl%26sysp...

ServiceNow All Favorites History Admin Access Control - New Record Search

Access Control New record Submit

Conditions

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2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role

bb1

Insert a new row...

Security Attribute Condition

Local or Existing ☒ Existing ☐ Local

Security Attribute

Condition All of these conditions must be met

-- choose field --

OR AND

The screenshot displays the 'Access Control - New Record' form in ServiceNow. The 'Security Attribute Condition' section is expanded, showing a 'Security Attribute' dropdown and a condition clause 'All of these conditions must be met'. Below this, the 'Criteria' section shows a condition 'Branch is EEE' with a 'Choose option' tooltip. The 'Advanced Condition' section is partially visible at the bottom.

The screenshot shows the ServiceNow 'Access Control - New Record' configuration page. The 'Condition' section has a filter: 'Branch is EEE'. The 'Script' section is set to 'Turn on ECMAScript 2021 (ES12) mode' and contains the following code:

```

1  function () {
7    if (gs.hasRole('bbi')){
8      return true;
9    }
10   // Deny access for all others
11   return false;
12 }();
13

```

5. Create the Create ACL

- Type: record, Operation: create, Name: u_institution_details
- Requires role: bb2

- No Data Condition needed here
- Click submit

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record

* Operation: create

Application: Global

Active: ☒

Advanced: ☐

Admin overrides: ☒

Decision Type: Allow If

Protection policy: -- None --

* Name: Institution Details [u_institution_details]

Description:

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

6. Create the Write ACL

- Type: record, Operation: write, Name: u_institution_details
- Requires role: bb3
- No Data Condition needed here
- Click Submit

ServiceNow Access Control - New Record

* Operation: write

Active: ☒

Admin overrides: ☒

Decision Type: Allow If

Protection policy: -- None --

* Name: Institution Details [u_institution_details]

Description:

Tables

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role

7. Create the Delete ACL

- Type: record, Operation: delete, Name: u_institution_details
- Requires role: bb4
- No Data Condition needed here
- Click Submit

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record
 * Operation: delete
 Application: Global
 Active: ☒
 Admin overrides: ☒
 Decision Type: Allow If
 Protection policy: -- None --
 * Name: Institution Details [u_institution_details]
 Description:
 -- None --

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

8. Verification Procedure

- Impersonate the test EEE user.
- Open the u_institution_details list.
- Verify the records visible to the user.
-
-

Access Controls

Impersonate user

Select a user

EEE User

EU EEE User

RECENT IMPERSONATIONS

Cancel Impersonate user

Name	Operation	Application	Active
\$allappsmgmt			
\$atf_page_inspector			
\$conversation-builder			
\$mycompanyappsmgmt			
\$products			
\$pwd_reset_serviceDesk			
\$sa_service_model_json			
\$sla_timeline			
\$sn-va-web-client-app			
\$sn_global_search_results			
\$spd			
\$spm_availability_calculation_setting			
\$studio			
\$upgrade_client			

-
- Impersonate an administrator and verify full access.
-
-

Script-Controlled ACL-Micro Pr...ServiceNow DevelopersInstitution Details | ServiceNow

dev446707.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do%3Fsysparm_clear_stack%3Dtrue

Ask Gemini

servicenow

AllFavoritesHistory

Institution Details

Branch

Search

Actions on selected rows...New

All

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Alfonso Griglen		Alejandra Prenatt	STU200

1 to 1 of 1

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